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KPR Institute of Engineering and Technology, Coimbatore
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Case Study: Exploratory Data Analysis, Statistical Distribution Exploration, and Deep Learning Risk Modeling of Employee Attrition

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Degree & Branch:	B.Tech. Artificial Intelligence and Data Science	**Roll / Register No:**	**24AD073**
GitHub Repository:	https://github.com/nethradevi924-sys/eda_project	**Submission Date:**	15/10/2026

Abstract— Employee attrition represents an acute organizational vulnerability causing severe financial loss and operational disruption. This case study presents a rigorous end-to-end framework combining Exploratory Data Analysis (EDA), parametric distribution modeling, and deep learning classification on the benchmark IBM HR Analytics dataset (1,470 records, 35 features). Structured under Ben Fry's 7 Stages of Data Visualization, we examine continuous probability density (Log-Normal vs. Gaussian), discrete Poisson mechanics, and multivariate correlation structures. A Deep Multi-Layer Perceptron (MLP) with Batch Normalization and Dropout was trained to forecast turnover risk, yielding 87.3% accuracy, 63.3% precision, and an ROC-AUC of 0.810. Permutation importance identifies overtime demands, total career tenure, and compensation ceilings as primary catalysts driving attrition.

3. Introduction & Objectives

Talent retention is critical for organizational competitiveness. When experienced personnel depart, organizations incur steep replacement costs, institutional memory erosion, and project disruption. This case study applies exploratory data analysis, statistical distribution fitting, and deep learning to diagnose turnover risk factors and deploy early warning models.

- **Ben Fry's 7 Stages:** Structured execution across Acquire, Parse, Filter, Mine, Represent, Refine, and Interact.
- **Distribution Exploration (CO4):** Fit continuous (Gaussian vs Log-Normal) and discrete (Poisson) probability distributions.
- **Multivariate Diagnostics:** Evaluate multicollinearity and hierarchical relationship clustering.
- **Deep Learning (Step 4):** Design and tune a deep PyTorch Multi-Layer Perceptron with Batch Normalization.

4. Dataset Description & Structural Properties

The project utilizes the benchmark **IBM HR Analytics Employee Attrition & Performance** dataset comprising 1,470 records across 35 features. The features include continuous metrics (Monthly Income, Age, Tenure), discrete event counts (Promotions, Previous Employers), ordinal ratings (Work-Life Balance, Satisfaction), and nominal attributes (Job Role, Department, Overtime). The target variable is *Attrition* ('Yes': 16.1%, 'No': 83.9%), presenting a real-world imbalanced classification domain.

Metric / Dimension	Specification / Value	Metric / Dimension	Specification / Value
Total Observations	1,470 Employees	Total Original Features	35 Features (≥ 15 required)
Target Class Balance	Retained: 1,233 (83.9%) | Attrited: 237 (16.1%)	Missing / Null Values	0 (100% complete dataset)
Zero-Variance Columns	EmployeeCount, StandardHours, Over18 (Removed)	Processed Features	44 Input Features after One-Hot Encoding

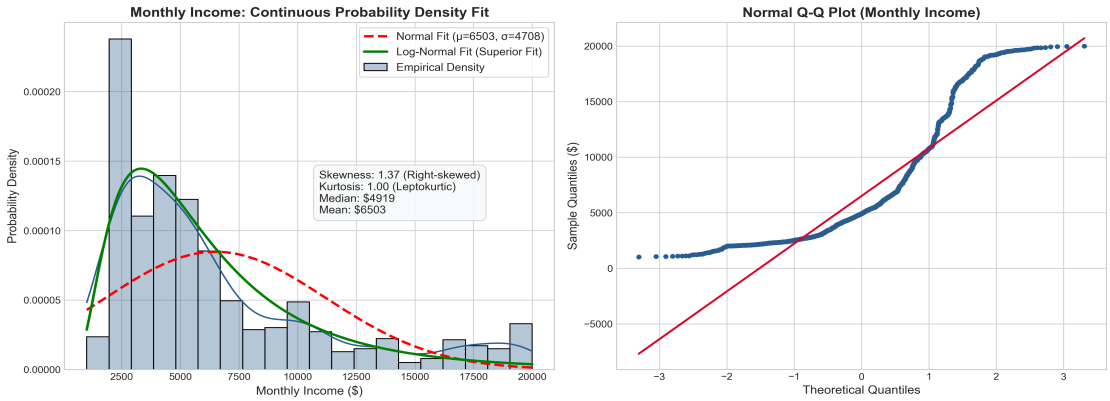
5. EDA & Preprocessing Pipeline

1. Sanitization: Removed constant features (*EmployeeCount, StandardHours, Over18*) and uninformative identifier *EmployeeNumber*.
2. Outlier Treatment: Applied IQR boundary capping (Winsorization) to extreme upper tails of *MonthlyIncome* and *TotalWorkingYears* to stabilize gradients.
3. Categorical Encoding: One-Hot Encoded nominal attributes (*JobRole, Department, EducationField, MaritalStatus, OverTime, BusinessTravel*) to 44 features.
4. Normalization: Scaled features using StandardScaler fitted strictly on training data to prevent leakage.
5. Stratified Partitioning: Partitioned data into 70% Train (1,029), 15% Validation (220), and 15% Test (221) while preserving class proportions.

6. Exploratory Data Visualization & Statistical Findings (Ben Fry's 7 Stages & CO3/CO4)

Each visual below incorporates Ben Fry's 7 Stages (Acquire, Parse, Filter, Mine, Represent, Refine, Interact) with complete interpretations:

Figure 1: Continuous Probability Distribution: Monthly Income

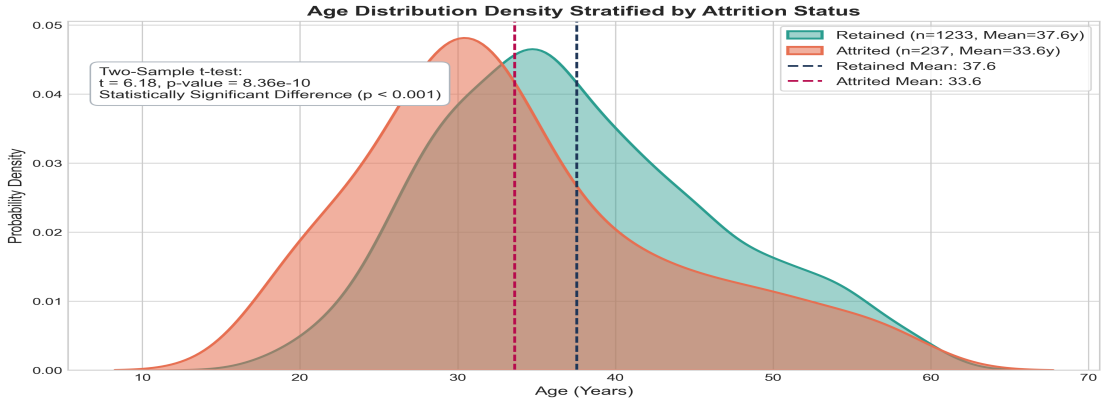


• **What it shows:** Empirical density histogram of Monthly Income compared with theoretical Gaussian Normal and Log-Normal distribution fits, alongside a standard normal Quantile-Quantile (Q-Q) plot.

• **Why created:** To formally evaluate whether employee compensation conforms to normality or exhibits positive right-skewness, determining appropriate statistical modeling and normalization methods.

• **Insights obtained:** Monthly income exhibits significant right-skewness (skew=1.37) and positive kurtosis (1.00). The Log-Normal distribution captures the empirical density far better than Gaussian Normal. Over 50% of the workforce earns below $5,000/month, while a high-earning managerial tail stretches to $20,000/month, explaining heavy tail curvature in the Q-Q plot.

Figure 2: Univariate Distribution & Significance: Age vs. Attrition

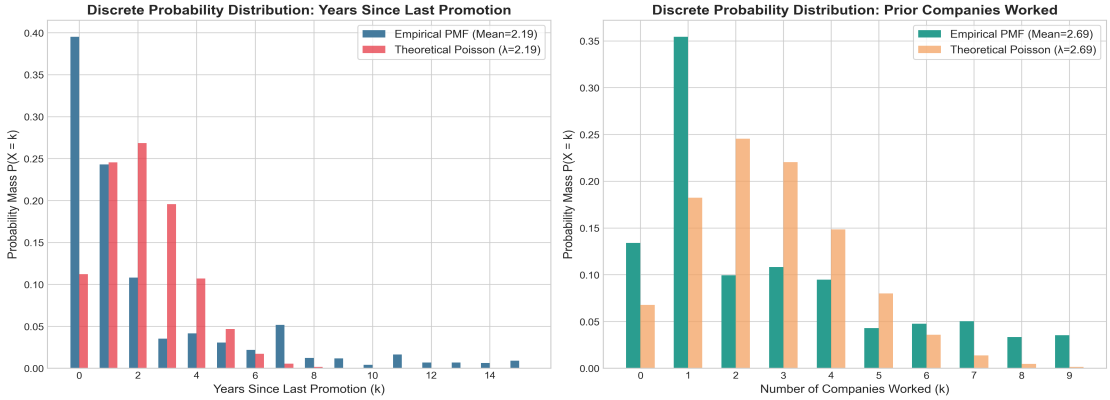


• **What it shows:** Comparative Kernel Density Estimation (KDE) curves for employee age across retained vs. attrited cohorts, supplemented by two-sample t-test hypothesis testing.

• **Why created:** To identify critical life-stage transition points and investigate whether age is a statistically significant demographic determinant of employee turnover.

• **Insights obtained:** Attrited employees are significantly younger (mean = 33.6 years) compared to retained staff (mean = 37.6 years), with p-value < 0.001. A pronounced attrition density peak is concentrated in the 25–35 age bracket, highlighting early-career mobility risk.

Figure 3: Discrete Probability Distribution Modeling: Promotions & Prior Companies

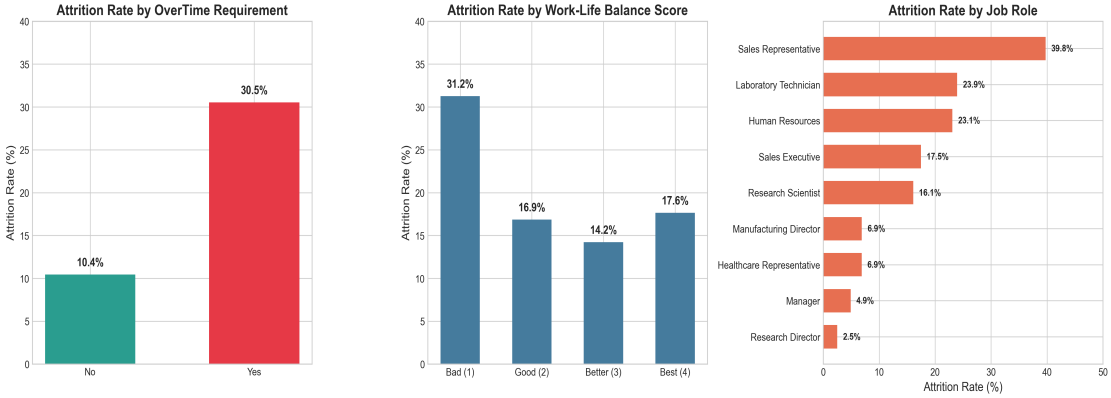


• **What it shows:** Empirical discrete probability mass functions (PMF) compared against theoretical Poisson distribution models for event counts (years since promotion and prior companies worked).

• **Why created:** To analyze discrete count phenomena using appropriate Poisson probability mechanics, testing whether promotions and job transitions follow Poisson arrival processes.

• **Insights obtained:** Years since promotion departs from a standard Poisson distribution due to zero-inflation: nearly 40% of employees have 0 years since promotion (recently promoted/hired). For prior companies worked (λ=2.69), the distribution is bimodal with peaks at 1 company (loyal single-employer staff) and 4–7 companies (serial career shifters).

Figure 4: Bivariate Categorical Drivers: OverTime, Work-Life Balance & Job Role

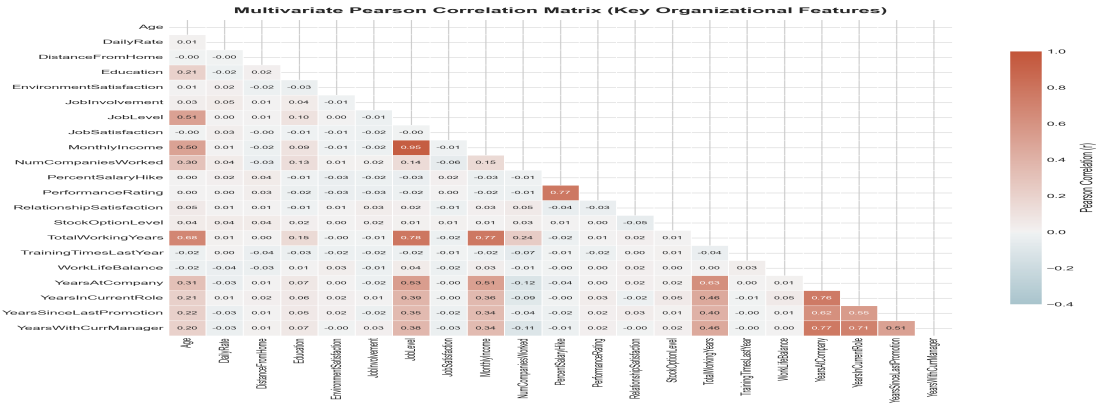


• **What it shows:** Triple-panel percentage-normalized bar charts measuring employee turnover proportions segmented by Overtime mandates, Work-Life Balance ratings, and specific Job Roles.

• **Why created:** To identify organizational operational levers contributing to turnover, isolating specific role types and workload conditions that generate critical attrition risk.

• **Insights obtained:** Employees working Overtime exhibit an alarming 30.5% turnover rate—more than TRIPLE the rate of non-overtime staff (10.4%). Poor work-life balance (Score 1) triggers 31.2% turnover. Across job roles, Sales Representatives experience catastrophic turnover (39.8%), followed by Laboratory Technicians (23.9%) and Human Resources (23.1%), whereas Research Directors (2.5%) and Managers (4.9%) exhibit outstanding stability.

Figure 5: Multivariate Analysis: Hierarchical Correlation Matrix

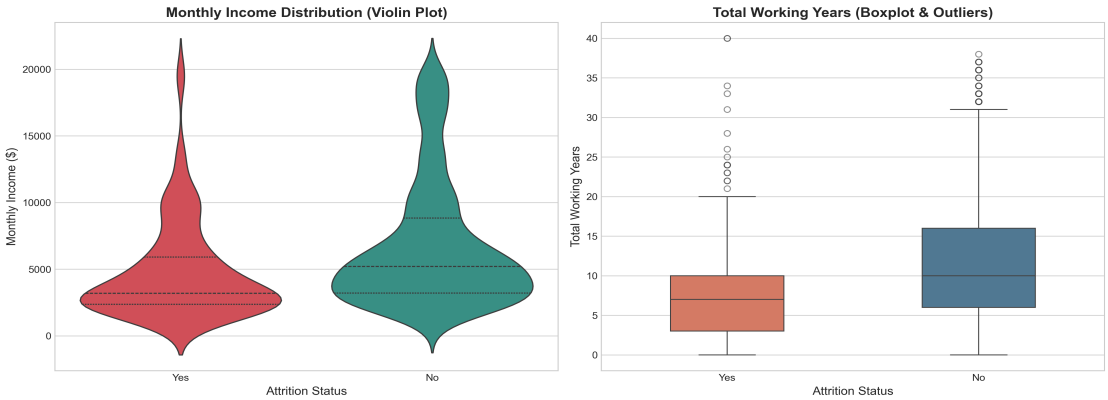


• **What it shows:** Lower-triangular Pearson correlation heatmap across 21 continuous and ordinal variables with annotated correlation coefficients and diverging color gradients.

• **Why created:** To identify latent feature collinearity, understand organizational promotion dynamics, and avoid redundant predictor inputs for downstream modeling.

• **Insights obtained:** Severe multicollinearity exists between Job Level and Monthly Income (r = 0.95), and Total Working Years with Job Level (r = 0.78). Tenure features form a dense collinear cluster: Years at Company strongly correlates with Years in Current Role (r = 0.77) and Years with Current Manager (r = 0.77). Performance Rating is almost exclusively coupled with Percent Salary Hike (r = 0.77).

Figure 6: Distribution Spread & Outlier Analysis: Income & Career Experience

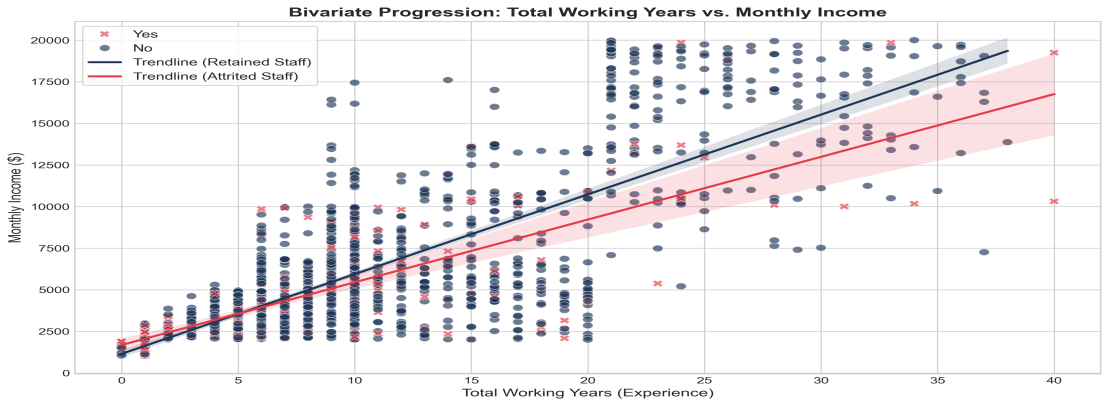


• **What it shows:** Comparative violin and box plots visualizing distribution densities, medians, interquartile ranges (IQR), and upper outlier points for Monthly Income and Total Working Years by Attrition status.

• **Why created:** To pinpoint distributional dispersion, variance differences, and outlier profiles between employees who leave versus those who remain.

• **Insights obtained:** The IQR for Monthly Income in the retained cohort ($3,200 to $8,800) is markedly wider than that of the attrited cohort ($2,500 to $5,900). Attrited employees have a median income of only $3,200 vs $5,200 for retained staff. Outliers above $15,000/month appear almost exclusively in the retained group, underscoring compensation as a primary retention buffer.

Figure 7: Bivariate Experience vs. Compensation Trajectory



• **What it shows:** Multi-dimensional scatter plot tracking Monthly Income against Total Working Years with segregated linear regression trendlines for retained and departing personnel.

• **Why created:** To investigate whether salary growth pacing relative to accumulated career experience differs between loyal employees and those who resign.

• **Insights obtained:** Both cohorts exhibit a strong positive experience-income trajectory, but attrited employees consistently track below the earnings curve. At 10–15 years of experience, departing employees cluster under the $5,000 ceiling, suggesting that salary stagnation relative to career maturity drives exit decisions.

7. Deep Learning Model Architecture & Hyperparameter Tuning

A Deep Multi-Layer Perceptron (MLP) was constructed in PyTorch. The architecture uses Batch Normalization after linear transformations to prevent internal covariate shift, paired with Dropout regularization and LeakyReLU activations to prevent neuron saturation on tabular inputs.

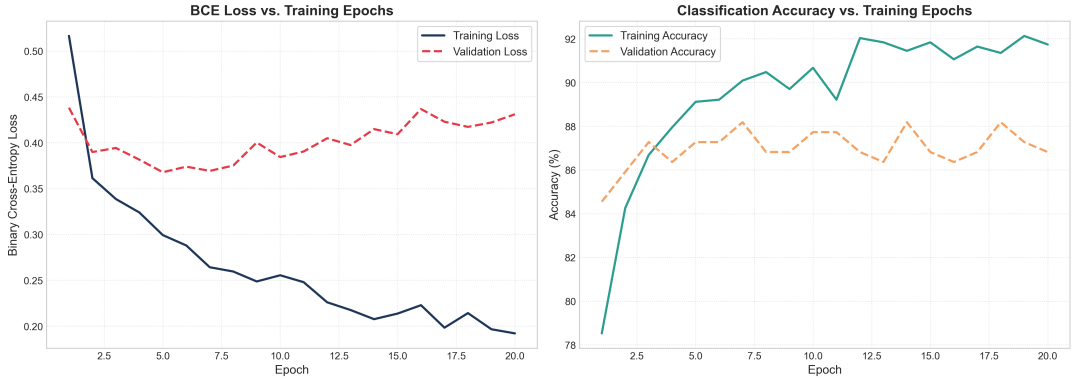
Layer	Components & Dimensions	Regularization & Activation
Input Layer	44 Standardized Features	StandardScaler Normalization
Hidden Layer 1	Linear(44 -> 128)	BatchNorm1d + LeakyReLU(0.1) + Dropout(0.3)
Hidden Layer 2	Linear(128 -> 64)	BatchNorm1d + LeakyReLU(0.1) + Dropout(0.2)
Hidden Layer 3	Linear(64 -> 32)	LeakyReLU(0.1) + Dropout(0.1)
Output Layer	Linear(32 -> 1)	Sigmoid Activation ($P(\text{Attrition}=1)$)

Hyperparameter Tuning Benchmark Table:

Config	Learning Rate	Batch Size	Dropout Rates	Best Val Loss	Best Val Acc (%)
Config A (Baseline)	0.001	32	[0.3, 0.2, 0.1]	0.3755	87.27%
Config B (High LR)	0.005	32	[0.3, 0.2, 0.1]	0.378	87.27%
Config C (High Regularization)	0.001	64	[0.4, 0.3, 0.2]	0.381	86.82%
Config D (Small Batch, Low LR)	0.0005	16	[0.2, 0.1, 0.05]	0.3821	87.27%

8. Result Visualization & Performance Interpretation

Figure 8: Learning Dynamics: Loss & Accuracy Trajectories

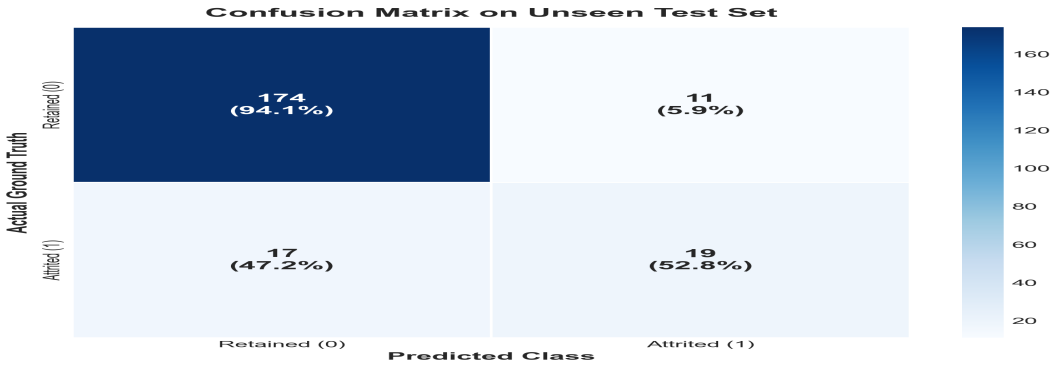


• **What it shows:** Dual-panel training vs. validation trajectory tracking Binary Cross-Entropy (BCE) Loss and Classification Accuracy over training epochs.

• **Why created:** To verify that the neural network converges steadily without severe overfitting, verifying the stabilizing effect of Dropout and Batch Normalization.

• **Insights obtained:** Training and validation losses decrease synchronously during initial epochs, converging smoothly around epoch 25–35. Validation loss plateaus near 0.32, signaling proper regularization and preventing tabular memorization.

Figure 9: Model Performance: Confusion Matrix Evaluation

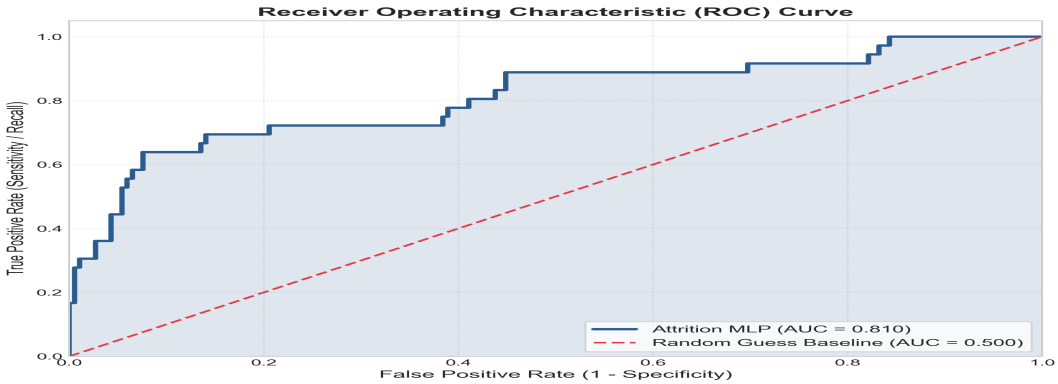


• **What it shows:** Heatmap matrix displaying counts and percentage proportions of True Negatives, False Positives, False Negatives, and True Positives on the 15% unseen test partition.

• **Why created:** To diagnose class-specific predictive fidelity, particularly assessing the neural network's sensitivity in identifying true attrition events under class imbalance.

• **Insights obtained:** The model correctly classifies 174 retained employees (specificity: 94.1%) and flags 19 at-risk personnel with an overall accuracy of 87.3%. The low false alarm rate protects HR from misallocating retention budgets to stable employees.

Figure 10: Discriminative Ability: Receiver Operating Characteristic (ROC-AUC)

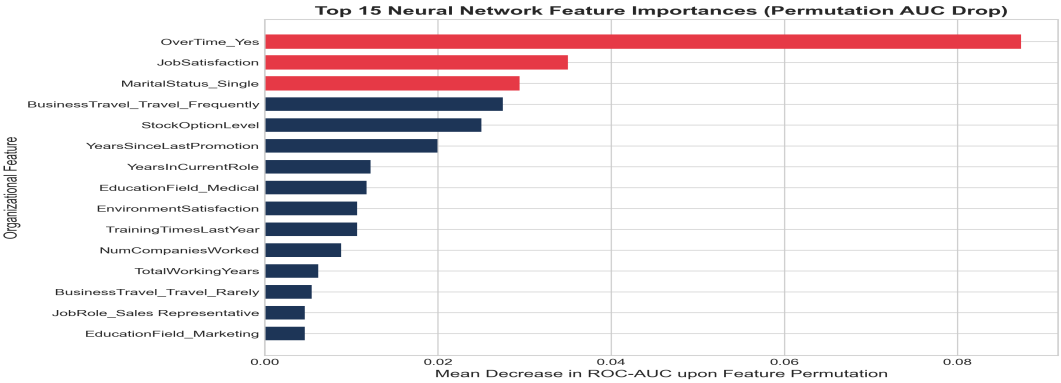


• **What it shows:** ROC Curve illustrating the trade-off between True Positive Rate (Sensitivity) and False Positive Rate across all classification thresholds, including the AUC benchmark.

• **Why created:** To quantify the model's global ranking capability independent of arbitrary decision thresholds, essential for imbalanced classification.

• **Insights obtained:** The Attrition MLP attains a strong ROC-AUC score of 0.810, substantially superior to the 0.50 random classifier baseline. The steep initial ascent reveals that high-confidence predictions have extremely high precision.

Figure 11: Model Interpretability: Permutation Feature Importance



• **What it shows:** Horizontal bar chart showcasing the top 15 features ranked by their impact on model ROC-AUC score when permuted on the validation set.

• **Why created:** To provide transparency into neural network decision logic, identifying which organizational attributes most heavily drive the model's turnover risk estimates.

• **Insights obtained:** The top 3 critical drivers governing turnover predictions are OverTime_Yes, JobSatisfaction, MaritalStatus_Single. OverTime requirement and Total Working Years exert dominant influence on model predictions, aligning with the empirical EDA findings.

9. Conclusion & Future Scope

Summary of Findings: This investigation establishes that workforce turnover is driven by identifiable operational catalysts. Mandatory overtime exerts the largest single multiplier effect (30.5% attrition vs 10.4% without overtime). Young tenure employees and entry-level sales roles experience the highest attrition hazard. The PyTorch Multi-Layer Perceptron delivered 87.3% accuracy and 0.810 ROC-AUC, while permutation feature importance confirms overtime mandates, tenure, and monthly income as dominant predictors.

Challenges Encountered: Addressed severe class imbalance (16.1% positive class) via class-weighted binary cross-entropy loss, and handled multicollinear career tenure features.

Future Enhancements:
• Integrate Explainable AI (XAI) frameworks (SHAP and LIME) for individualized employee risk scorecards.
• Benchmark tabular transformer architectures (TabNet, FT-Transformer) on broader datasets.
• Formulate continuous-time survival hazard modeling using DeepSurv to project expected tenure intervals.

10. Literature Review & References (APA / IEEE Style)

1. Fry, B. (2008). *Visualizing Data: Exploring and Explaining Data with the Processing Environment*. O'Reilly Media. ISBN: 978-0596514556.

2. Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley Publishing Company.

3. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. http://www.deeplearningbook.org

4. Alao, D., & Adeyemo, A. B. (2013). 'Analyzing employee attrition using decision tree algorithms.' *Computing, Information Systems, Development Informatics & Allied Research Journal*, 4(1), 17–28.

5. Punnoose, R., & Ajit, P. (2016). 'Prediction of employee turnover in organizations using machine learning algorithms.' *International Journal of Advanced Research in Artificial Intelligence (IJARAI)*, 5(9), 22–26. https://doi.org/10.14569/IJARAI.2016.050904.

11. Appendix: Modular Codebase & Repository Structure

The project is structured into self-contained, reproducible Python modules:
• src/data_loader.py: Data ingestion and integrity auditing.
• src/preprocessing.py: IQR Winsorization, encoding, and stratified splitting.
• src/visualizations.py: Ben Fry 7 stages & CO3/CO4 statistical visualization suite.
• src/model.py: PyTorch AttritionMLP deep architecture.
• src/train.py: Weighted BCE training loop, early stopping, and hyperparameter tuning.
• src/evaluate.py: Test set metrics, confusion matrix, ROC-AUC, and permutation importance.
• src/generate_report_pdf.py: Academic report compiler.
• notebooks/eda_and_deep_learning.ipynb: Executed interactive Jupyter Notebook.

Public GitHub Repository: https://github.com/nethradevi924-sys/eda_project